

Assignment

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Course code :- ITA1416

Course :- ETHICAL HACKING

2. objectives :-

- To identify active host within the university network.
- To perform non-destructive ICMP ping sweeps
- To identify TCP and UDP services using Nmap.
- To enumerate services running on windows and linux
- To identify potentially vulnerable or unnecessarily exposed services.

3. scope of assignment :-

Component	Assessment
Student web portal	web vulnerability assessment.
windows application server	Port /service and configuration assessment.
windows administrative server	SBM, remote administration configuration assessment.
Linux database server	SSH, database daemon assessment.
Linux file server	NFS, Permission service assessment.

Assignment :- Penetration testing and cybersecurity assessment of a university smart campus infrastructure.

Introduction :-

Modern universities depends heavily on digital infrastructure for academic, administrative financial and research activities.

A smart campus environment may contain student web portals, application servers, wireless network file-sharing services, remote administration terms and various network-connected services.

The assessment covers :-

- network reconnaissance.
- Host discovery and ICMP ping sweeps
- TCP and UDP port scanning
- service enumeration
- windows server security
- linux server security
- web application security.
- vulnerability analysis.

5. dataset selection & attack pattern analysis.

for analysis portion, a suitable publicly available cybersecurity dataset is the CICISD 2017 dataset

- Port scanning
- Brute force authentication
- web application attacks.
- denial of service behaviour.
- Infiltration & lateral movement
- CIA Traid Impact Analysis.
- TCP port scanning.
- UDP port scanning
- hping3 assessment.
- Service enumeration
- web application penetration testing.
- Nikto assessment.
- Burp suite assessment

Broken access control:-

ex:- student → password ✓

student → faculty password ✗

student → administration panel ✗

included

out of scope

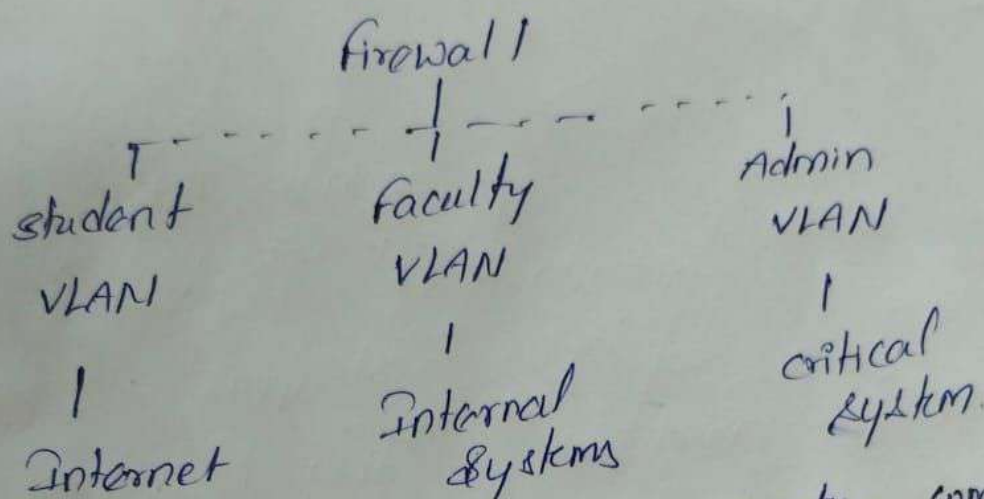
- Testing real university systems.
- destructive exploitation.
- denial of service attacks.
- data destruction.
- password theft from real users.

4. Laboratory Environment

System	Example IP	Operating System	Purpose
Kali Linux	192.168.56.10	Kali Linux	Attacker's Machine
Web server	192.168.56.20	Linux	Student portal
Windows server	192.168.56.30	Windows server	Application server
Admin server	192.168.56.31	Linux	Database
File server	192.168.56.41	Linux	NFS File server
Wireless AP	lab controlled	lab configuration	wireless

wireless network segmentation.

wireless networks should be separated according to user type.



student devices should not directly communicate with admin servers.

Conclusion :-

The penetration testing assessment demonstrates how a university smart campus environment can be systematically evaluated using ethical hacking methodologies.

The assessment begins with host network followed by TCP & UDP & scanning and service enumeration.